1 Introduction

In modern research on science and engineering, experimental and simulation-based studies increasingly rely on high-dimensional data to study complex physical systems [1, 2]. Despite growing data volumes, discoveries are often difficult to reproduce, compare, or transfer across experimental and computational settings [3, 4]. A key limitation is that many data-driven workflows do not explicitly encode symmetry, i.e., the transformations under which a system exhibits equivalent behavior, so physically equivalent observations are treated as distinct. This leads to redundant parameterizations, fragile models, and limited generalization beyond narrow conditions. **The goal of this POSE Phase I project** is to address this gap by transitioning `PyDimension` [5] from a successful but distributed set of open-source research tools into a unified Open Scientific Ecosystem (OSE), called *OpenSymmetry*. *OpenSymmetry* provides a shared, reproducible framework for data-driven symmetry discovery, enabling physically interpretable knowledge extraction directly from data. Rather than introducing new learning algorithms, the project focuses on organizing, validating, and sustaining symmetry-aware discovery workflows that are currently scattered across disciplines. This transition lays the foundation for community-driven development, consistent validation, and reuse of symmetry-based discoveries across scientific and engineering domains.

What Is Symmetry in Scientific Discovery and Why It Matters. Symmetry describes invariance of a system under transformations that do not change its observable behavior. A simple example is a square, which remains unchanged under rotations by 90 degrees. More broadly, symmetries constrain how physical systems can vary. Their importance is well established in physics, including the 1957 Nobel Prize–winning work of Chen Ning Yang and Tsung Dao Lee, which showed that parity symmetry can be violated in weak nuclear interactions [6]. A central theoretical result is Noether's theorem [7], which links continuous symmetries to conservation laws. In engineering, similar ideas appear through dimensional analysis. The Buckingham Π theorem [8] identifies scaling symmetries among variables and is widely used to reduce parameter spaces and uncover transferable relationships. When applied to data, identifying symmetry means recognizing transformations under which different measurements represent the same underlying phenomenon [9–13]. This reveals low-dimensional structure, removes redundant degrees of freedom, and yields representations that better reflect intrinsic system behavior.

Demonstrated Impact and Community Reach. `PyDimension` is a validated open-source library for data-driven discovery of symmetry structures in high-dimensional systems (Figure 1). It provides an end-to-end workflow for identifying scaling and translational symmetries directly from data, enabling systematic dimension reduction and physically interpretable representations. The library was released with two Nature Communications publications [14, 15], where it reliably rediscovered classical dimensionless numbers and scaling laws, including the Rayleigh number in Rayleigh–Bénard convection, laser–material interaction processes, and porosity prediction in advanced manufacturing. ❶ *Community Reach:* Since its release, `PyDimension` has been widely adopted by independent research groups across multiple scientific and engineering communities, including fluid dynamics and transport processes [16, 17]; materials science and manufacturing [18–20]; solid and applied mechanics [21, 22]; biomedical and chemical systems [23]; and cyber-physical and engineered systems [24–26]. These citations include over 10+ publications in high-impact venues such as Science and Nature, indicating recognition beyond a single application domain. ❷ *Educational Impact:* `PyDimension` supports education and workforce training through a well-documented workflow, Dimensionless Learning 2.0 tutorials [27], and an interactive web-based interface [28]. These materials have been used in invited seminars, including the CMU Scientific Machine Learning (ML) Webinar [29] and the "Math + ML + X" seminar at Brown University [30].

Impact on National Interest. `PyDimension` has demonstrated direct relevance to national priorities in advanced manufacturing and defense by enabling reproducible and physically interpretable analysis of complex manufacturing processes. Using experimental data from U.S. Department of Energy (DOE) national laboratory facilities, the library was used to discover a universal dimensionless number governing keyhole dynamics in metal additive manufacturing [15]. This result won the first prize in the AM Bench 2022 Challenge Awards hosted by the National Institute of Standards and Technology (NIST) [31], providing independent validation of both scientific significance and practical relevance. The development of `PyDimension` is funded and supported by the U.S. Army DEVCOM Analysis Center (DAC). Beyond individual studies, `PyDimension` has been referenced in DOE- and NIST-affiliated technical reports [32]

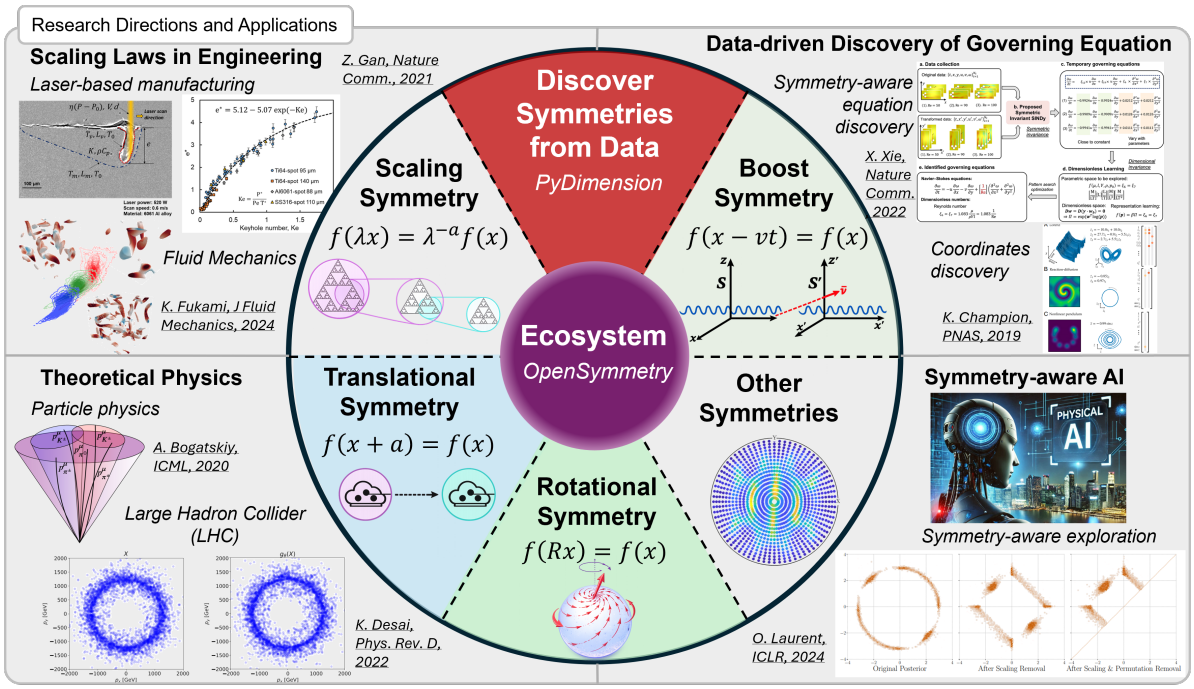

Figure 1: *OpenSymmetry* ecosystem for data-driven symmetry discovery and its applications. The figure illustrates a unified ecosystem centered on discovering symmetries directly from data (via `PyDimension`) and organizing them into fundamental symmetry classes, including scaling, translational, rotational, boost, and other symmetries.

and strategic planning documents [33] as a representative example of data-driven, physics-consistent analysis for advanced manufacturing and scientific ML. These reports emphasize the role of dimensionless and symmetry-aware approaches in reducing data requirements, improving generalization, and enabling reusable scientific models. Such alignment indicates that the methods underlying `PyDimension` address capability gaps identified in national research infrastructure and manufacturing technology roadmaps [34].

Ecosystem Vision. The current library includes complete modules for scaling and translational symmetries built on a well-designed extensible framework. The proposed *OpenSymmetry* aims to enhance distributed development of additional symmetry modules, such as rotational and boost symmetries, expanding our user base from dozens to hundreds, and the contributor base from a few to several dozen. The proposed OSE targets four groups of users and contributors: ❶ Experimentalists from engineering and science who seek hidden symmetry and structure in their data. ❷ Physicists and mechanicians working on data-driven discovery of governing equations. `PyDimension` will interface with existing differential equation discovery libraries to enable symmetry-aware equation discovery. ❸ AI scientists who want to ensure that their generative AI models respect fundamental symmetry constraints. ❹ Educators and mathematicians who need accessible toolboxes for learning and teaching symmetry and group theory. This structure allows users to reproduce results, compare methods under consistent assumptions, and extend to new domains.

Landscape of Existing Tools and Gaps. A growing set of open-source tools supports components of data-driven scientific discovery, including symbolic regression libraries (`PySR` [35]), sparse identification (`PySINDy` [36]), and physics-informed ML (`DeepXDE` [37]). However, these tools lack systematic and standardized workflows for enforcing or discovering symmetry structures, which limits interpretability, extrapolation, and reproducibility. Recently developed group-theory-based libraries (`LieGAN` [38] and `LieSD` [39]) focus primarily on mathematical symmetry discovery, lacking direct interfaces to real physical systems, such as dimensional analysis, and they are not designed to interoperate with the existing discovery approaches described above. These gaps motivate the *OpenSymmetry*, built on `PyDimension`.

Team Qualification. The project team brings combined expertise in physical modeling, applied mathematics, and open-source software development. **PI Gan** has a strong track record in multiphysics modeling and simulation, data-driven equation discovery, and open-source project management. He is the creator and maintainer of `PyDimension`, as well as JAX-AM [40] and JAX-FEM [41]. **Co-PI Wei** has rich experience

in creating and maintaining open-source tools that are widely adopted by academia, industry, and government. He is the active maintainer of several open-source projects [42–45] with over 1000+ stars/forks on GitHub. He has led tutorials [46, 47] and data challenges [48] that introduced open-source tools to a broad audience of researchers and practitioners. The PI and Co-PI also have on-going collaboration on AI for science [49]. The project will also engage **J. Zhang**, a senior software architect at Microsoft, as an OSE mentor. She brings extensive experience in building and sustaining large distributed software systems.

1.1 Intellectual Merit

OpenSymmetry advances fundamental knowledge in data-driven scientific discovery by establishing a reproducible, symmetry-aware framework for discovering scaling laws and governing structures directly from data. Rather than introducing new learning algorithms, the project formalizes how symmetry-aware discovery is integrated, validated, and sustained as shared scientific infrastructure. ❶ *Integration and reproducibility of symmetry-aware discovery. OpenSymmetry* defines a canonical end-to-end workflow that integrates dimensional reasoning, invariance constraints, and system identification within a unified framework, enabling reproducible comparison and transfer of symmetry-based discoveries across physical domains. ❷ *Distributed governance models for scientific open-source infrastructure.* The project develops and evaluates contribution workflows and governance structures tailored to interdisciplinary scientific software, advancing understanding of how rigor, physical consistency, and reproducibility can be maintained in distributed open-source ecosystems. ❸ *Collaborative benchmarking and validation practices.* Shared benchmarks, modular extension interfaces, and execution pipelines support systematic evaluation of symmetry discovery methods under noisy data, and experimental variation, connecting theory with reproducible reuse.

1.2 Broader Impacts

OpenSymmetry will generate impacts across education, community participation, and national research by: ❶ *Lowering barriers for experimental scientists and engineers.* An open and reproducible workflow enables researchers to extract physically interpretable structure from sparse or noisy data without reliance on opaque, domain-specific modeling assumptions. ❷ *Broadening participation through education and training.* Public documentation, tutorials, and interactive tools support students and early-career researchers in learning symmetry-aware and physics-informed data analysis, contributing to interdisciplinary workforce development. ❸ *Advancing national priorities in manufacturing, energy, and trustworthy AI.* Reproducible discovery of symmetries and governing structures aligns *OpenSymmetry* with federal priorities in advanced manufacturing, effective use of national research infrastructure, and trustworthy AI for scientific applications.

2 Context of OSE

2.1 Vision and Guiding Principles

OpenSymmetry will serve as the managing organization for a community-driven open-source ecosystem that enables reproducible, physically grounded scientific discovery through data-driven symmetry and dimensionless learning. The long-term vision is to make the discovery of symmetry invariants, scaling laws, and governing equations from experimental and computational data *reproducible*, *comparable*, and *transferable* across physical domains, platforms, and institutions. This enables scientific knowledge extracted from sparse and noisy datasets to be trusted, reused, and extended beyond individual studies.

The guiding principles are: *(i) Integration-first*, aligning core `PyDimension` modules for dimensional specification and symmetry determination into a single canonical end-to-end symmetry discovery workflow; *(ii) interoperability*, through stable, domain-agnostic interfaces that allow different symmetry classes to interoperate with downstream equation discovery tools such as `PySINDy` and `DeepXDE`; *(iii) reproducibility*, supported by shared benchmarks from validated `PyDimension` applications, including Rayleigh–Bénard convection and Keyhole Number discovery, together with versioned artifacts and containerized workflows; *(iv) transparency*, enabled by open governance and clear documentation that make specifications, assumptions, and limitations explicit to users; and *(v) sustainability*, ensured by lightweight governance structures, interface stability guarantees, and distributed maintenance practices beyond any single research group. These principles map directly to the project aims: integration-first design underlies the canonical workflow in Sec 3.1; interoperability and reproducibility enable distributed development and validation in Sec 3.2; and transparency and sustainability guide governance and community formation in Sec 3.3.

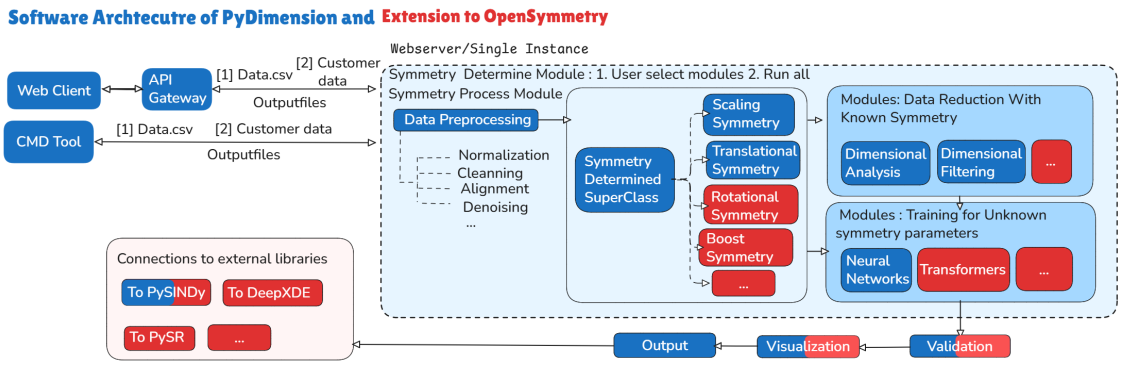

Figure 2: Software architecture of current `PyDimension` framework (blue) and its evolution toward *OpenSymmetry* (red), showing a modular symmetry discovery pipeline with plug-in symmetry types and shared interfaces that support reproducibility and extensibility.

2.2 Current Product and Community Status

Existing Foundations and Open-Source Product. *OpenSymmetry* builds on `PyDimension`, which organizes modular components for dimensional specification, symmetry-aware optimization, statistical filtering, and visualization, and has been validated through Dimensionless Learning for discovering scaling and translational symmetries in applications such as fluid dynamics and metal additive manufacturing. The codebase is publicly available under a permissive license and is accompanied by documentation [50], tutorials [27], and an interactive web-based interface [28] designed for experimental scientists and engineers.

Development, Testing, and Dissemination. `PyDimension` currently follows standard open-source development practices (Figure 2), including public issues, pull-request–based codes, and tagged releases. Reference workflows and example notebooks support reproducible execution of published results, and tutorial materials provide structured guidance for new users. While these practices support research use and education, they do not yet provide organizational structure for sustained, distributed ecosystem development.

Limitations of Current and Opportunity The original version of `PyDimension` focused on data-driven dimensional analysis via dimensionless learning. This approach embeds the Buckingham Π theorem [8] into a machine learning framework, which captures universal scaling invariants in physical systems. We later recognized that the dimensionless learning is mathematically equivalent to scaling symmetry in parameter space [14] or to translational symmetry in log-parameter space [51]. This insight led us to re-design a broader framework that extends beyond scaling and translational symmetries to include more general symmetry structures. Despite demonstrated research and educational impact, the current framework remains a collection of successful research artifacts rather than a managed open-source ecosystem. First, symmetry discovery workflows are not standardized end to end. Results often depend on domain-specific scripts and assumptions, which limits reproducibility and comparison across datasets and application areas. Second, there are no formal pathways for external contributors to add new symmetry classes, benchmark datasets, or discovery modules in a consistent manner. Third, the absence of a governance structure, interface stability guarantees, and coordinated release practices creates risks for long-term maintenance and reuse as adoption grows. *OpenSymmetry* addresses these limitations by transitioning the existing codebase into a community-driven open-source ecosystem, consistent with the goals of POSE Phase I.

2.3 *OpenSymmetry* as an Ecosystem and Its Novelty

OpenSymmetry addresses these challenges by transitioning `PyDimension` into a managed open-source ecosystem centered on a canonical workflow for data-driven symmetry discovery. From a user perspective, the ecosystem provides a coherent and repeatable process: users supply data and symmetry or dimensional specifications, select compatible discovery modules, and obtain reproducible and physically interpretable symmetry structures, including scaling laws and governing equations. The novelty of *OpenSymmetry* lies in three aligned elements. ❶ *Canonical symmetry discovery workflow. OpenSymmetry* will define a stable, end-to-end workflow for symmetry discovery that spans data ingestion, symmetry specification, invariance enforcement, and model recovery. Dimensionless Learning will be treated as one symmetry class within this workflow, alongside other transformation groups, allowing results to be reproduced and compared

across domains. ❷ *Distributed development through shared interfaces and benchmarks.* The ecosystem will provide clear extension points for adding new datasets, physical systems, and symmetry discovery modules without modifying core infrastructure. Shared benchmarks derived from existing applications will support consistent evaluation under varying noise levels, data sparsity, and material or system variation. ❸ *Governance and sustainability foundations. OpenSymmetry* will be supported by a managing organization with a clear charter, roadmap, release and deprecation policies, and basic security practices. Governance and long-term sustainability are reinforced by external partners and adopters documented through letters of support, including Brookhaven National Laboratory and the U.S. Army DEVCOM Analysis Center, which validate sustained use of `PyDimension` on national laboratory and government datasets; industrial research participation from Meta scientist X. Xie and international academic adoption from Italy Politecnico di Torino by A. Ciampaglia. Together, these partners provide complementary perspectives on scientific validity, applied use, and ecosystem growth.

3 Project Development Plan

As identified in Sections 1 and 2, the primary barriers to broader adoption of data-driven symmetry discovery are not algorithmic, but structural: the lack of a canonical end-to-end workflow, the absence of standardized contribution and validation pathways, and the need for lightweight but durable governance to support distributed development. Consistent with the goals of POSE Phase I, this project focuses on ecosystem formation and readiness with three connected aims (Figure 3).

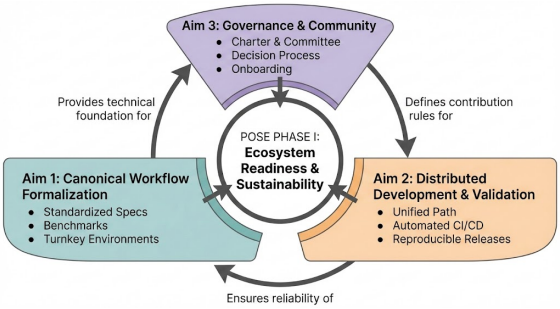

Figure 3: Synergies among three aims

3.1 Aim 1: Formalization of a Canonical Symmetry Discovery Workflow

This aim addresses the lack of a standardized, end-to-end workflow for data-driven symmetry discovery identified in Section 2. The goal is to formalize how existing symmetry-aware components in `PyDimension` are composed, specified, and validated within a single reproducible workflow.

Subtask 1.1: Canonical Symmetry Discovery Workflow. We will define a canonical workflow that explicitly specifies the sequence and interfaces for (i) data ingestion and preprocessing, (ii) dimensional and symmetry specification, (iii) symmetry-constrained optimization, and (iv) reconstruction of interpretable models such as scaling laws or reduced governing equations. The workflow will formalize the inputs and outputs of each stage (e.g., datasets, dimensional matrices, symmetry generators, and validated coefficients), decoupling discovery logic from ad-hoc scripts across implementations.

Subtask 1.2: Reference Benchmarks and Validation Tasks. To make the workflow reproducible, we will curate a small set of reference benchmarks drawn from established uses of `PyDimension`. Initial benchmarks will include canonical fluid dynamics cases (e.g., recovery of Reynolds and Rayleigh scaling symmetries) and additive manufacturing case studies (e.g., discovery of the Keyhole Number). Each benchmark will package the dataset, symmetry or dimensional specification, expected discovery outputs, and explicit validation criteria, serving as unit tests for end-to-end reproducibility.

Subtask 1.3: Turnkey Execution Environments. The canonical workflow will be packaged into turnkey execution environments, including Docker distributions and Google Colab notebooks. Each release will bundle the workflow specification, benchmarks, validation scripts, and a minimal example that produces expected results without manual configuration.

Deliverables. Deliverables for Aim 1 include: (i) a published canonical workflow specification with documented interfaces; (ii) a reference benchmark suite with reproducible reports; and (iii) containerized and cloud-ready execution environments. Aim 1 will be considered successful if independent users can understand the workflow and validate benchmark without assistance from the development team.

3.2 Aim 2: Distributed Development and Continuous Validation

This aim responds directly to the lack of standardized contribution and validation pathways identified in Section 2. The objective is to enable distributed contributions while preserving dimensional consistency,

symmetry invariance, and end-to-end reproducibility.

Subtask 2.1: Unified Contribution Path. We will define a single contribution pathway for all changes to the ecosystem, including new benchmark datasets, symmetry modules (e.g., scaling or translational symmetry components), workflow extensions, and documentation. All contributions will be submitted through a hub repository accompanied by standardized metadata describing the dataset or module, its symmetry or dimensional assumptions, and expected outputs. Each contribution will follow the same review, validation, and merge process, reducing ad hoc or undocumented changes.

Subtask 2.2: Automated Validation and Regression Checks. Automated validation will be implemented to explicitly check (i) dimensional consistency of inputs and learned representations, (ii) preservation of specified symmetry or invariance constraints, and (iii) reproducibility of reference benchmark results. Pull requests will trigger lightweight checks on a small benchmark subset, while scheduled runs will execute a broader benchmark suite to detect regressions under noise, subsampling, or configuration changes. Validation outcomes will be published as compact reports linked to each contribution and release.

Subtask 2.3: Reproducible Releases and Contributor Recognition. Regular, versioned releases will bundle the workflow specification, benchmark suite, validation reports, and reference results. Contributors will be acknowledged in release notes, and changes to core interfaces or specifications will require additional review to preserve backward compatibility and ecosystem stability.

Deliverables. By the end of Aim 2, *OpenSymmetry* will operate a hub repository with a unified contribution path, automated validation, and reproducible, versioned releases. Aim 2 will be considered successful if independent contributors can add benchmarks or modules through the unified workflow and reproduce published results in a clean environment without core-team intervention.

3.3 Aim 3: Governance and Community Growth

This aim establishes the governance and community practices required for *OpenSymmetry* to function as a sustainable open-source ecosystem beyond Phase I. The focus is on piloting transparent mechanisms for decision-making, accountability, and onboarding, rather than building a fully mature governance structure.

Subtask 3.1: Governance Structure and Charter. A small Governing Committee will be established to oversee workflow standards, benchmark inclusion, and releases (Figure 4). The committee will operate under a concise charter defining roles, decision procedures, conflict-of-interest rules, and a public request-for-comment process for major changes. Membership will span multiple institutions and disciplines.

Subtask 3.2: Accountability and Sustainability Practices. Lightweight practices will be adopted, including public documentation of decisions, tracking of usage and contribution metrics, and security and maintenance procedures. These practices support continuity as the ecosystem grows beyond the founding team.

Subtask 3.3: Community Onboarding and Outreach. Structured on-ramps will be created through targeted outreach and training. The project will offer short tutorial sessions and short courses at conferences TMS (San Diago), WCCM (Munich), and KDD (Jeju) to engage experimentalists, computational scientists, and AI for science communities. To support asynchronous learning, the team will produce a small series of YouTube introduction and tutorial videos [52]. Additional engagement will include one or two virtual workshops, a quarterly community call, and cross-posted examples illustrating interoperability with complementary tools such as PySINDy.

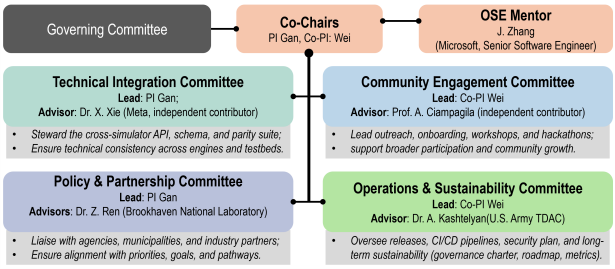

Figure 4: Organization and governance structure.

Deliverables. Deliverables for Aim 3 include a published governance charter, documented contribution processes, onboarding materials, and a public summary of community activity. Aim 3 will be considered successful if external contributors participate in releases and governance activities during Phase I.

4 Risk Analysis / Security Plan

Transitioning `PyDimension` into *OpenSymmetry* introduces risks specific to reproducible, symmetry-aware scientific software, where correctness depends on preserving dimensional and symmetry constraints. Phase I

Task & Time		Lead	Q1	Q2	Q3	Q4	Primary Indicators	Success Criteria
Aim 1	1.1: Unified Integration Backbone	Gan					(i) Canonical workflow spec + stable interfaces. (ii) Compact benchmarks (Rayleigh–Bénard, Keyhole). (iii) Containerized / cloud-ready bundles.	(i) ≥2 external groups reproduce end-to-end results. (ii) Expected symmetry recovered across environments. (iii) Fresh install → result ≤ 30 min; no core-team help.
	1.2: Parity Suite	Gan						
	1.3: Easy-to-use Environments	Wei						
Aim 2	2.1: One Contribution Path	Gan					(i) Unified contribution hub with standardized templates and review flow. (ii) Automated validation checks for dimensional consistency, symmetry invariance, and reproducibility. (iii) Regular versioned releases with published validation reports.	(i) Median time-to-first-response on external pull requests ≤ 5 business days. (ii) ≥ 90% of automated validation runs complete and publish reports. (iii) ≥ 3 releases during Phase I, with ≥ 2 including external contributions.
	2.2: Automated Validation Checks	Gan						
	2.3: Releases and Recognition	Wei						
Aim 3	3.1: Committee and Charter	Gan					(i) Governing Committee with charter, roles, and public decision records. (ii) Documented policies for releases, interfaces, and maintenance. (iii) Onboarding materials and regular community engagement activities.	(i) ≥ 1 completed request-for-comment (RFC) cycle with a recorded decision. (ii) ≥ 10 first-time external contributors. (iii) ≥ 6 institutions successfully reproduce at least one benchmark.
	3.2: Accountability and Sustainability	Gan						
	3.3: Growth and Outreach	Wei						
Aggregated OSE Readiness		Gan					(i) Reproducible workflows, automated validation, and active governance. (ii) Evidence of external reproduction, extension, and community.	(i) ≥ 2 independent groups reproduce results; ≥ 1 extends the workflow without core-team help. (ii) ≥ 1 completed RFC with recorded decision; all NSF reports submitted on time with public artifacts.
NSF Reports		Gan						

Figure 5: Phase I work plan for *OpenSymmetry*: timeline of tasks, deliverables, and evaluation metrics across.

focuses on managing risks from distributed development, data contributions, and increasing use of AI-assisted coding tools. ❶ **Software Integrity, Reproducibility, and AI-Assisted Coding Risks.** AI-assisted code contributions may introduce subtle violations of dimensional or symmetry constraints that are not detected by standard tests. Phase I will require protected branches, pull-request review, and continuous integration checks that explicitly validate dimensional consistency, symmetry invariance, and benchmark reproducibility for all contributions, including AI-assisted ones. Releases will be versioned and distributed with containerized execution environments and an SBOM. ❷ **Data Integrity and Provenance Risks.** Symmetry discovery is sensitive to data quality and provenance. Phase I will enforce data contribution guidelines that document provenance, permissions, and preprocessing steps, and restrict benchmarks to synthetic, anonymized, or publicly releasable datasets. ❸ **Scientific Validity and Misinterpretation Risks.** Incorrect symmetry specifications or overreliance on automated discovery may lead to invalid conclusions. Phase I emphasizes validation by pairing benchmarks with expected outcomes, documented assumptions, and known failure modes, and by distinguishing validated results from exploratory findings.

5 Evaluation and Work Plan

Evaluation Plan. Phase I evaluation combines aim-specific criteria with an overall assessment of *OpenSymmetry*'s readiness as an open-source ecosystem. All workflow specifications, benchmark definitions, validation reports, governance records, and release artifacts will be publicly available. For **Aim 1**, evaluation focuses on integration and reproducibility. Success is measured by whether the same dataset and symmetry specification can be processed through the canonical workflow to reproduce validated results, such as recovery of known scaling symmetries, without ad hoc intervention. For **Aim 2**, evaluation focuses on the maturity of distributed development, including automated validation, contribution workflows, and reproducible releases. For **Aim 3**, evaluation focuses on governance transparency and early community participation (Figure 5). To assess **ecosystem readiness**, we will publish a quarterly scorecard aggregating four dimensions: *workflow reproducibility*, *development maturity*, *governance transparency*, and *community traction*. Phase I will be considered successful if at least two independent groups reproduce one benchmark using the packaged environment, extend the workflow with a new dataset or module without core-team assistance, and complete at least one request-for-comment (RFC) cycle with a documented decision.

Work Plan and Milestones. Quarterly tasks, deliverables, and evaluation milestones are summarized in Figure 5. In *Q1*, we will establish the *OpenSymmetry* integration hub, draft the canonical workflow specification, and publish the initial governance charter. In *Q2*, we will release Workflow Specification v0.1, the initial benchmark suite with validation reports, and a containerized execution environment, followed by an onboarding workshop. In *Q3*, we will conduct cross-site reproducibility tests, expand automated validation, and complete the first RFC cycle. In *Q4*, we will consolidate results through a second community sprint, a public evaluation dashboard, and a Phase II readiness package including updated specifications, benchmark evidence, governance records, and community metrics.

6 Results from Prior NSF Awards

PI Gan has not received prior NSF funding.
Co-PI Wei is the PI on ***(a)*** NSF award #2442477 "CAREER: The Next Frontier in Urban Data Analytics: From Prescriptive to Actionable", $599,999.00; 07/15/25-06/30/30. ***(c) Intellectual Merit:*** The project deals with sim-to-real challenges in urban environments. ***Broader Impacts:*** The results will support sustainable urban development. ***(d) Publications***: N/A. ***(e) Evidence of research products:*** N/A.